
Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

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Abstract

We examine the reported association between feature-importance concentration and conformal coverage degradation in a COVID-era supply-chain case study and external datasets. Rounded historical summaries yield Spearman $\rho = 0.853$ for a selected 16-task subset and $\rho = 0.654$ for all 17 reported tasks. A statistical audit identified test-aware preprocessing, reuse of calibration labels for early stopping, different prediction-set conventions across cohorts, and missing numerical artifacts. These findings limit the original pre-deployment and replication claims. A recovered, separate follow-up panel of all eight SALT tasks (three fixed seeds per task) gives $\rho = -0.238$ under inclusive score inversion and -0.262 under crossing sets. Its frozen predictions and SHAP concentrations were independently regenerated from raw inputs and stored models, and all 24 follow-up fits were rerun. This panel does not reproduce the historical positive association; the original 50-seed models remain unavailable. We give a conditional score bound for an idealized probability mixture, whose weight is distinct from measured SHAP concentration, and withdraw unsupported class-count and ICC interpretations. The diagnostic and 40% threshold are unvalidated; no deployment rule is supported by these results.

1 INTRODUCTION

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [Vovk et al., 2005]. However, real-world deployments face distribution shifts that violate this assumption. While prior work characterizes *how* conformal prediction degrades under shift [Tibshirani et al., 2019, Barber et al., 2023], a criti-

cal gap remains: **Can we identify which deployed models will fail before observing test data?**

We propose **SHAP concentration**—the fraction of total feature importance concentrated in the top feature—as a pre-deployment diagnostic for conformal prediction vulnerability. Using a supply chain benchmark with temporal splits aligned to COVID-19 (February–July 2020), we conduct an observational case study where 8 classification tasks experience common temporal periods yet exhibit coverage drops ranging from negligible ($<1\%$) to catastrophic ($>70\%$).

Our contributions:

1. **Exploratory diagnostic:** The reported SHAP concentration summaries correlate with coverage-drop summaries: $\rho = 0.853$ across the selected 16 tasks and $\rho = 0.654$ across all 17 tasks including Stack Overflow. These correlations can be recomputed from rounded published points, but the underlying experiment ledgers are missing and historical SALT/external set conventions differ. The recovered eight-task follow-up panel does not reproduce the positive association (Section 8). Prospective diagnostic value remains unestablished.
2. **Conditional theory:** Theorem 1 bounds an idealized probability mixture with explicit assumptions and a weight distinct from measured SHAP concentration. It does not verify those assumptions empirically or establish a guarantee for the archived experiments. Historical shift-detector comparisons are descriptive and await reproducible outputs.
3. **Exploratory framework:** A 40% concentration rule and protective-factor check summarize observed cases. They are retrospective hypotheses, with one external false negative and a protective rule derived from one false positive. Recovered single-seed retraining logs mix validation and test months; neither they nor the non-replicating follow-up panel validate a deployment recommendation.

2 RELATED WORK

Conformal Prediction. Vovk et al. [2005] introduced conformal prediction with exchangeability guarantees; Shafer and Vovk [2008] provide an accessible tutorial. Recent work extends to classification [Romano et al., 2020, Cauchois et al., 2021] and regression [Romano et al., 2019, Lei et al., 2018]. Angelopoulos and Bates [2023] provide a comprehensive survey. Ding et al. [2023] show that class-conditional coverage guarantees become harder to achieve as class count grows and propose clustered conformal prediction as a remedy; Gibbs et al. [2025] provide finite-sample conditional guarantees via a covariate-shift reweighting framework interpolating between marginal and conditional validity. Both works frame class count as a challenge—a confound we explicitly examine via partial correlation analysis (Section 5.4).

Conformal Prediction Under Shift. Tibshirani et al. [2019] study covariate shift with known propensity scores. Podkopaev and Ramdas [2021] develop distribution-free uncertainty quantification for classification under label shift. Barber et al. [2023] provide a comprehensive treatment beyond exchangeability, bounding coverage loss by the total variation distance between test and calibration score distributions. Kasa and Taylor [2023] empirically characterize how CP degrades across many vision architectures and shift types, showing substantial degradation is common—our work addresses the complementary question of which tabular models will fail *before* test data is observed. We complement these theoretical characterizations with an empirical diagnostic that is associated with this distributional divergence—and hence with which tasks fail—before deployment.

Adaptive Methods. Gibbs and Candès [2021, 2024] develop Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [2022] for time series, Feldman et al. [2023] for online risk control, and Bhatnagar et al. [2023] for strongly adaptive online learning. Angelopoulos et al. [2024] generalize conformal prediction to broader risk measures. Kasa et al. [2025] propose entropy-based methods (ECP/EACP) that adapt prediction sets using unlabeled test data at deployment time—a related but distinct goal from ours, which requires no test observations: SHAP concentration is a structural pre-deployment diagnostic computed solely on validation data. Our experiments show ACI recovers marginal coverage under severe shift but at a substantial cost in prediction set informativeness.

Shift Detection. WILDS [Koh et al., 2021] and Shifts [Malinin et al., 2021] provide benchmarks, while Gulrajani and Lopez-Paz [2021] show domain generalization methods often fail to improve over ERM under realistic distribution shifts. Gardner et al. [2023] benchmark tabular distribution shift across 15 tasks—the closest tabular-specific bench-

mark; our SALT dataset differs in providing a naturalistic COVID-19 temporal split with documented business-disruption context rather than curated benchmark partitions. Garg et al. [2022] use unlabeled test data to predict accuracy degradation—a related goal but requiring test-time observations. Kernel-based tests (MMD [Gretton et al., 2012]) and classifier two-sample tests (C2ST [Lopez-Paz and Oquab, 2017]) detect distributional differences but do not predict the *severity* of downstream failure. We show empirically that MMD and C2ST detect shift for all tasks uniformly yet cannot distinguish catastrophic from robust conformal outcomes (Section 6.3). Our contribution is a *pre-deployment* diagnostic using feature importance structure that predicts failure severity, not merely shift presence. Miller et al. [2021] show that in-distribution (ID) validation accuracy predicts out-of-distribution (OOD) accuracy for image classifiers—a correlation that does not hold for conformal coverage in our tabular setting, where ID validation coverages cluster near the 90% target, spanning 83.6%–99.9% (Table 2), yet test-time coverage spans 12–100%, reinforcing the need for structure-based diagnostics beyond accuracy metrics.

Interpretability for Reliability. Our work connects SHAP [Lundberg and Lee, 2017, Lundberg et al., 2020] to model reliability assessment, extending feature attribution from post-hoc analysis to prospective failure diagnosis. Our use is prospective: concentration is computed pre-deployment on validation data, predicting which models will fail.

3 METHODOLOGY

3.1 DATASET AND TEMPORAL SPLITS

We use the SALT (Supply chain ALlocATion) dataset from RelBench [Robinson et al., 2024], a supply chain benchmark with temporal splits. All 8 classification tasks from the rel-salt benchmark are included; no tasks were excluded. External analyses report 9 additional datasets. We retain the historical $K \geq 4$ subset (8 external datasets plus 8 SALT tasks, $n = 16$) as an exploratory scope choice. APS mechanics do not rule out catastrophic coverage failure at $K = 2$ or 3, and no dated prospective exclusion plan is available. Including Stack Overflow ($K = 3$) yields the all-17 sensitivity result $\rho = 0.654$, $p = 0.004$; this sensitivity limits conclusions from the selected subset.

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

Audit of the historical data protocol. The supplied SALT code fits categorical encoders and the target vocabulary using train, validation, and test data. It also uses all valida-

tion labels for early stopping before dividing that sample into calibration and evaluation; the external script likewise reuses validation labels. Thus, although the reported SHAP summaries use validation covariates, the full pipeline is not strictly pre-deployment and does not satisfy standard held-out split-conformal assumptions. The numerical tables below are historical reported results, not a validated rerun. Corrected evaluation requires train-only preprocessing, an independently defined class vocabulary, and separate tuning/calibration/evaluation samples. Threshold evaluation uses test-drop labels and is retrospective.

3.2 CONFORMAL PREDICTION SETUP

The archived experiments use two deterministic APS-related conventions. SALT set construction includes the class that crosses the calibrated cumulative-probability threshold. The external script instead computes coverage by the event $s(x, y) \leq \hat{q}$, while its size calculation includes the crossing class. These define different sets; the historical pooled endpoint and external coverage/size pairs therefore require reevaluation under one explicit convention. A new reference implementation uses the exact finite-sample order statistic and inverts a fixed-tie-order inclusive APS score. It has not generated the historical results reported here.

1. Train LightGBM classifier with fixed hyperparameters
2. Calibrate conformal predictor on 50% of validation set
3. Evaluate coverage on held-out validation and test sets

3.3 ENSEMBLE PROTOCOL

The historical design reports 50 seed runs (42–91) on the same data, with a fixed chronological validation split, t intervals across seeds and paired Wilcoxon comparisons. These summarize algorithmic variation conditional on one dataset and do not create 50 independent realizations of a distribution shift. Original seedwise results are unavailable for recomputing the intervals/tests.

3.4 FEATURE OVERLAP METRIC

Feature temporal stability uses Jaccard similarity:

$$J(x_j) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where A_{train} and A_{test} are the sets of unique values for feature x_j in train and test data. In the pre-deployment framework (Section 7), we substitute validation data for test data as a proxy.

3.5 SHAP CONCENTRATION DIAGNOSTIC

We define the SHAP concentration metric as:

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j} \quad (2)$$

where ϕ_j is the mean absolute SHAP value of the j -th feature computed on validation data using TreeExplainer [Lundberg et al., 2020], and ϕ_1 is the importance of the top-ranked feature. C is computed **before deployment** using only validation data.

3.6 WHY SHAP CONCENTRATION?

A natural question is whether simpler diagnostics suffice.

Feature overlap (Jaccard) is insufficient. Tasks relying on transaction IDs have Jaccard ≈ 0 while tasks with entity-based features have Jaccard > 0.5 (Table 3), yet Jaccard alone cannot explain the full range of coverage drops.

Post-hoc metrics require test data. Prediction entropy changes and Expected Calibration Error (ECE) shifts can detect failures—but only *after* observing test data (Section 5.4). Among the diagnostics we evaluate, SHAP concentration is available before deployment and shows the strongest severity association.

Why not raw feature importance? Raw importance values are not comparable across tasks (different scales, feature counts). Concentration normalizes to a $[0, 1]$ scale, enabling cross-task comparison.

Why top-1? Top-1 has the strongest observed association among the reported concentration summaries. The table reports eight candidate metrics (top-1/2/3/5/10, HHI, Gini, entropy). With the displayed SALT points, its unrounded two-sided Spearman $p = 0.0101755$ gives Holm-adjusted $p = 0.0814$ for this eight-test family (0.0509 even for the previously reported five-test family). Neither clears 0.05. The pooled $n = 16$ result reuses these eight SALT tasks and is not independent replication of metric selection. On the eight external rounded summaries alone, $\rho = 0.833$, $p = 0.0102$; this is descriptive arithmetic, pending provenance and pipeline reconciliation.

Model scope. The reported model-sensitivity correlations differ (LightGBM 0.833, CatBoost 0.667, XGBoost 0.548, RF 0.30, MLP 0.43). These observational comparisons do not identify architectural causes or establish that boosting necessarily reinforces single-feature dependence, bagging necessarily dilutes it, or SHAP approximation error explains the MLP result. TreeSHAP is exact for the specified tree-explanation model, but a causal mechanism would require controlled interventions and attribution-stability checks. Generalization beyond the evaluated models remains unverified.

Table 1: Per-Task Feature Importance Concentration. Top- k shows cumulative importance in the top k features (%). Drop = coverage degradation (pp). High Top-1 with catastrophic drop (e.g., s-payterms: 54.2% $\rightarrow$ 77.1 pp) versus high Top-2/3 with near-zero drop (s-office: 77.0%/84.8% $\rightarrow$ 0.1 pp) is an observed contrast, not evidence of a causal mechanism.

Task	Top-1	Top-2	Top-3	Drop
s-payterms	54.2	78.1	89.6	77.1
s-shipcond	50.7	77.5	92.5	71.6
i-shippoint	48.8	72.7	92.4	18.5
s-group	47.3	65.9	79.6	71.2
s-office	42.6	77.0	84.8	0.1
i-incoterms	28.9	42.3	55.1	11.3
i-plant	23.9	46.9	65.0	10.6
s-incoterms	23.7	47.2	69.1	8.5

Tasks sorted by Top-1 concentration. Top- k = $\sum_{j=1}^k \phi_j / \sum_{j=1}^p \phi_j$ (%). Drop = mean coverage drop (50 seeds, pp). Spearman ρ with drop: Top-1 = 0.833 ($p = 0.010$ before multiplicity correction); Top-2 = 0.524 ($p = 0.183$); Top-3 = 0.524 ($p = 0.183$); Top-5 = 0.690 ($p = 0.058$); Top-10 = 0.399 ($p = 0.328$). Alternative concentration metrics: HHI = 0.619 ($p = 0.102$); Gini = 0.500 ($p = 0.207$); Entropy = 0.571 ($p = 0.139$)—none significant. The contrast for sales-office motivates a mechanism hypothesis; the stability of individual features is not established by this correlation comparison.

4 THEORY: WHY CONCENTRATION RELATES TO FAILURE

We give a conditional score bound for an idealized probability mixture. Its weight λ is distinct from measured SHAP concentration C . The calculation is not a guarantee for the archived experiments and does not establish their mechanism.

Score and set convention. Fix a deterministic class-index tie order. Let $s(x, y^*)$ be the sum of probabilities through the true label in decreasing-probability order, and let $\mathcal{B}(x, y^*)$ contain the labels ranked after it. Then

$$s(x, y^*) = 1 - \sum_{y \in \mathcal{B}(x, y^*)} \hat{p}(y | x). \quad (3)$$

Each omitted probability is at most $\hat{p}(y^* | x)$, including ties. The score-inversion set is $\mathcal{C}_q(x) = \{y : s(x, y) \leq q\}$. This convention differs from the historical SALT crossing-label set; that distinction must be resolved before using this proposition to interpret its coverage.

Theorem 1 (Conditional bound for an idealized probability mixture). *Let $K \geq 2$, $\lambda \in [0, 1]$, and a fixed deterministic threshold $q \in [0, 1]$. Assume*

- (A1) $\hat{p}(y | x) = \lambda g(y | x_1) + (1 - \lambda)h(y | x_{\setminus 1})$, where g, h are probability distributions.
- (A2) At test time $g(y^* | x_1) \leq \varepsilon$ almost surely, for fixed $\varepsilon \in [0, 1]$.

- (A3) The joint distribution of $(h(\cdot | x_{\setminus 1}), y^*)$ is the same in calibration and test populations.

Write $\bar{h} = \mathbb{E}[h(y_{\text{cal}}^* | x_{\setminus 1}^{\text{cal}})]$. Then

$$s(x^{\text{test}}, y^*) \geq 1 - (K - 1)\{\lambda\varepsilon + (1 - \lambda)h(y^* | x_{\setminus 1}^{\text{test}})\}, \quad (4)$$

and consequently

$$\mathbb{E}[s_{\text{test}}] \geq 1 - (K - 1)\{\lambda\varepsilon + (1 - \lambda)\bar{h}\}. \quad (5)$$

Holding ε and $\bar{h}$ fixed, this lower bound is strictly increasing in λ if $\varepsilon < \bar{h}$. This concerns the bound, not monotonicity of the actual score expectation.

For $\lambda < 1$, score-inversion coverage obeys

$$\mathbb{P}(y^* \in \mathcal{C}_q(x^{\text{test}})) \leq \mathbb{P}(h(y_{\text{cal}}^*) \geq T(\lambda)), \quad (6)$$

where

$$T(\lambda) = \frac{(1 - q)/(K - 1) - \lambda\varepsilon}{1 - \lambda}.$$

Holding q, ε and the distribution of $h(y_{\text{cal}}^*)$ fixed, this upper bound is non-increasing when $\varepsilon \leq (1 - q)/(K - 1)$; the threshold is strictly increasing under strict inequality. At $q = 1, \varepsilon = 0$ it is constant. At $\lambda = 1$, if $q < 1 - (K - 1)\varepsilon$ coverage is zero; otherwise this argument yields only the trivial upper bound one.

Proof. The omitted sum in (3) contains at most $K - 1$ terms, each at most $\hat{p}(y^*)$. Substituting (A1)–(A2) gives (4); expectation and (A3) give (5). For the score-inversion event, $s \leq q$ implies the lower bound is at most q , yielding (6) when $\lambda < 1$. Its threshold derivative is $[(1 - q)/(K - 1) - \varepsilon]/(1 - \lambda)^2$. The $\lambda = 1$ case follows directly from the pointwise bound. $\square$

The proposition does not identify λ with the mean-absolute TreeSHAP ratio, whose usual LightGBM decomposition is in raw-margin space. Recalibrating a changed classifier also changes q and may change h , so the fixed-quantity monotonicity does not establish monotone empirical coverage across classifiers. The historical score comparisons in Appendix H are illustrative substitutions, not verification of (A1)–(A3).

5 RESULTS

5.1 COVERAGE DEGRADATION ACROSS TASKS

Despite experiencing common COVID-era temporal periods, the 8 supply chain tasks exhibit dramatically different coverage degradation (Table 2). The historical code uses a one-sided paired Wilcoxon alternative for positive drops and reports $p \leq 0.005$, with t intervals over seeds. Without the seed ledger these tests cannot be verified. Reported mean drops range from 0.1% to 77.1%.

Table 2: Coverage Degradation Under COVID-19 Distribution Shift (50 Seeds). 95% CIs from t -distribution. p -values from paired Wilcoxon signed-rank test (val $\rightarrow$ test, seed-level). All drops significant at $p \leq 0.005$.

Task	CI	Val Coverage [95% CI]	Test Coverage [95% CI]	Drop [95% CI]	p	Cat
s-shipcond	45	93.5 [93.2, 93.7]	21.8 [14.1, 29.6]	71.6 [63.9, 79.4]	<0.001	SEV
s-group	459	83.6 [81.7, 85.5]	12.4 [3.1, 21.7]	71.2 [61.3, 81.0]	<0.001	SEV*
s-payterms	137	90.8 [90.7, 91.0]	13.7 [6.0, 21.5]	77.1 [69.4, 84.9]	<0.001	SEV
i-plant	35	92.0 [91.7, 92.3]	81.4 [79.0, 83.8]	10.6 [8.2, 13.0]	<0.001	ROB
i-shippoint	69	91.2 [90.9, 91.4]	72.7 [62.3, 83.0]	18.5 [8.2, 28.8]	0.005	ROB* [†]
s-incoterms	13	95.5 [95.3, 95.7]	87.0 [84.7, 89.3]	8.5 [6.2, 10.8]	<0.001	ROB
i-incoterms	13	95.0 [94.9, 95.2]	83.7 [80.9, 86.6]	11.3 [8.5, 14.1]	<0.001	ROB
s-office	25	99.9 [99.9, 100.0]	99.9 [99.9, 99.9]	0.1 [0.0, 0.1]	<0.001	ROB

SEV = Severe (>50% drop), ROB = Robust (<20% drop); *high model variance (CV > 50%), classified by median; s-group’s reported 83.6% validation coverage is below target; high class count alone does not explain failure of a marginal conformal guarantee. Calibration reuse, chronological shift and implementation require investigation; [†] meets At-risk criterion (drop >15 pp) under the decision framework (Table 11).

Table 3: Feature Overlap for Primary Features.

Task	Primary Feature	Jaccard	Drop
s-shipcond	SALESDOC (ID)	0.02	71.6%
s-group	SALESDOC (ID)	0.02	71.2%
i-incoterms	PRODUCT, PARTY	0.58	11.3%
s-office	SALESORG	0.61	0.1%

Drop = mean val – mean test coverage (50 seeds).

5.2 DIAGNOSTIC ANALYSIS

Two factors account for much of the variance:

Factor 1: Feature temporal stability. Tasks using transaction IDs (Jaccard ≈ 0.02) fail catastrophically; tasks using stable entities (products, organizations; Jaccard > 0.5) maintain coverage (Table 3).

Factor 2: Feature importance concentration. Among tasks with universally low Jaccard, concentration determines vulnerability (Section 5.3).

5.3 SHAP CONCENTRATION CORRELATES WITH FAILURE

Across all 8 SALT multiclass tasks, SHAP concentration correlates with coverage degradation: Spearman $\rho = 0.833$, $p = 0.010$ (Table 4, Figure 2). A newly rerun percentile task bootstrap of the rounded summaries gives [0.29, 1.00] (10,000 resamples, seed 42), versus the historical report [0.30, 1.00]. Treating shared-domain tasks as exchangeable resampling units is not justified by this calculation; it is a sensitivity summary, not an established population interval. A BCa interval has not been computed. The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though statistical significance is not maintained for 2 of 8 jackknife samples ($p = 0.052$) due to reduced power at $n = 7$. SHAP concentration values are

highly stable: bootstrap resampling (1,000 resamples of 10K validation samples) yields coefficient of variation below 1% for all tasks, with 95% CIs within ± 1 percentage point.

Mechanism. Figure 1 reveals the distinction. Catastrophic tasks: the dominant feature experiences a concentrated importance explosion ($4.5\times$ increase). Robust tasks: importance redistributes across features with ranking reshuffle ($10\text{--}20\times$ increases distributed). These observations do not verify the probability-mixture assumptions of Theorem 1.

False positives. Under the mean-drop >15 pp criterion, sales-office is the SALT false positive; item-shippoint is at risk by mean although most seeds have small drops. The proposed protective rule was derived from sales-office alone, and inconsistent descriptions of its feature identity and overlap split require the original feature ledger. No deployment guarantee follows from this retrospective rule.

Dependence and uncertainty. SALT tasks share entities, periods and one benchmark. Their statistical independence is not established by between-task ICC. Seed-level intervals describe algorithmic variability conditional on these data; they do not quantify uncertainty across new datasets or future shifts. Negative validation/test agreement ICC cannot establish negative correlation or conservative paired tests (Appendix G).

Inferential status. Both the within-SALT and pooled cross-domain analyses are exploratory. The pooled endpoint includes the SALT tasks used in diagnostic development. Observed-effect power calculations do not confer confirmatory status, and no dated prespecified analysis plan or raw prediction ledger is available in the released package.

Cross-domain extension. The pooled historical summaries yield $\rho = 0.853$ and Kendall $\tau = 0.667$ for the selected 16 tasks. The supplied eight external rounded points alone yield $\rho = 0.833$, $p = 0.0102$, while all 17 tasks yield $\rho = 0.654$, $p = 0.004$. These arithmetic checks do not validate the underlying missing runs, independence of task selection, or

Table 4: Stratified Correlation Analysis.

Group	n	ρ	Kendall τ	$p(\rho)$	Boot. 95% CI
Multiclass (SALT)	8	0.833	0.714	0.010	[0.30, 1.00]
Multiclass (4 dom.)	11	0.909 [†]	0.782	<0.001	[0.61, 1.00]
Multiclass (8 dom.)	15	0.882 [‡]	0.714	<0.001	[0.60, 0.97]
Multiclass (9 dom.)	16	0.853	0.667	<0.001	[0.50, 0.96]
Combined (10 dom.) [§]	17	0.654	—	0.004	—

LOO stability (SALT): $\rho \in [0.75, 0.96]$; 6/8 jackknife significant.

Multiclass (9 domains) is the primary result, excluding Stack Overflow under the historical scope choice. Under the at-risk criterion (drop > 15 pp), threshold performance at 40% is Precision = 0.83, Recall = 0.83 (Appendix D). [†]Single-seed external values; multi-seed-consistent value at $n = 11$ is $\rho = 0.818$, $p = 0.002$ (see Appendix G). [‡] $n = 15$ is an intermediate analysis (7 external multiclass domains); $n = 16$ adds KDDCup99 as the 8th external domain to complete the 9-domain primary endpoint. Stack Overflow (3 classes, excluded scope subset) is excluded from all multiclass endpoints.

[§]Combined row includes all 17 tasks (8 SALT + 9 external, 10 domains: SALT + 9 external), adding Stack Overflow back to the $n = 16$ primary endpoint. Spearman $\rho = 0.654$, $p = 0.004$ (from Appendix G); Kendall τ not reported for this auxiliary endpoint.

comparability of historical APS conventions. Covertypes is the reported catastrophic external case and KDDCup99 a low-concentration false negative.

5.4 DIAGNOSTIC COMPARISON

We compare SHAP concentration against two alternative pre-deployment diagnostics that require no test data (Table 5).

Native feature importance concentration: top-1 LightGBM gain divided by total gain (zero SHAP cost, single seed; seed=42). Achieves $\rho = 0.667$ ($p = 0.071$)—directionally correct but not significant at $\alpha = 0.05$. The weaker discrimination arises because native gain conflates feature cardinality with predictive importance: sales-office shows 66.1% native concentration (vs. 42.6% SHAP) despite near-zero coverage drop.

Ensemble disagreement: standard deviation of validation coverage across 50 seeds. Achieves $\rho = 0.452$ ($p = 0.260$). Disagreement captures model instability but not the feature-level mechanism driving conformal failure.

Class cardinality as a possible confound. The historical within-SALT partial correlations are non-significant (0.629, $p = 0.131$ for concentration; 0.334, $p = 0.464$ for log K). The reported pooled partial correlation is 0.771 ($p = 0.0008$) for concentration and -0.010 ($p = 0.97$) for log K . These are conditional associations, not evidence that concentration causally drives severity independently of model, domain, shift magnitude, or pipeline differences. Their numerical provenance and uncertainty require the original task/seed ledger.

Table 5: Pre-Deployment Diagnostic Comparison ($n = 8$). Spearman ρ with 50-seed mean coverage drop. ρ_{partial} : partial Spearman controlling for the other variable in the SHAP–class-count pair.

Diagnostic	ρ	p	ρ_{partial}	p_{partial}
SHAP concentration	0.833	0.010	0.629	0.131
log(num_classes)	0.743	0.035	0.334	0.464
Native FI concentration	0.667	0.071	—	—
Ensemble disagreement	0.452	0.260	—	—

We additionally compare against two *post-hoc* baselines computed from 10-seed ACI experiments:

Prediction entropy change: $\Delta H = H_{\text{test}} - H_{\text{val}}$, where $H = -\sum_c p_c \log p_c$ averaged over instances.

Expected Calibration Error change: $\Delta \text{ECE} = \text{ECE}_{\text{test}} - \text{ECE}_{\text{val}}$.

Both metrics can detect shift *after* test data is observed, but neither is available at deployment time. SHAP concentration is computed solely on validation data, making it actionable for pre-deployment risk assessment. Comparison across 7 tasks in Appendix F.

6 EXTENDED EXPERIMENTS

6.1 ADAPTIVE CONFORMAL INFERENCE ACROSS ALL TASKS

The historical clipped ACI variant reports coverage improvements on severe tasks with larger sets. It clips α_t to $[0.001, 0.999]$ and quantile levels to $[0, 1]$, so the original arbitrary-shift ACI guarantee does not automatically transfer. It assumes immediate labels in the processed test order. Missing prediction ledgers prevent rechecking the reported tradeoff, and a deployability judgment requires task-specific utility and feedback timing.

6.2 ALTERNATIVE CONFORMAL SCORING

The historical Regularized Adaptive Prediction Sets (RAPS) [Angelopoulos et al., 2021] summaries report task-dependent differences: substantially reduces drops for high-cardinality tasks (s-group: 73.5% $\rightarrow$ 10.4%; s-payterms: 79.9% $\rightarrow$ 35.1%) but not the most concentrated task (s-shipcond: 60.4% $\rightarrow$ 67.8%). These differences do not establish the idealized mechanism in Section 4; its mixture weight is not measured SHAP concentration (Appendix J; RAPS used random 50/50 calibration splits vs. deterministic main-experiment splits—within-experiment comparisons are unaffected).

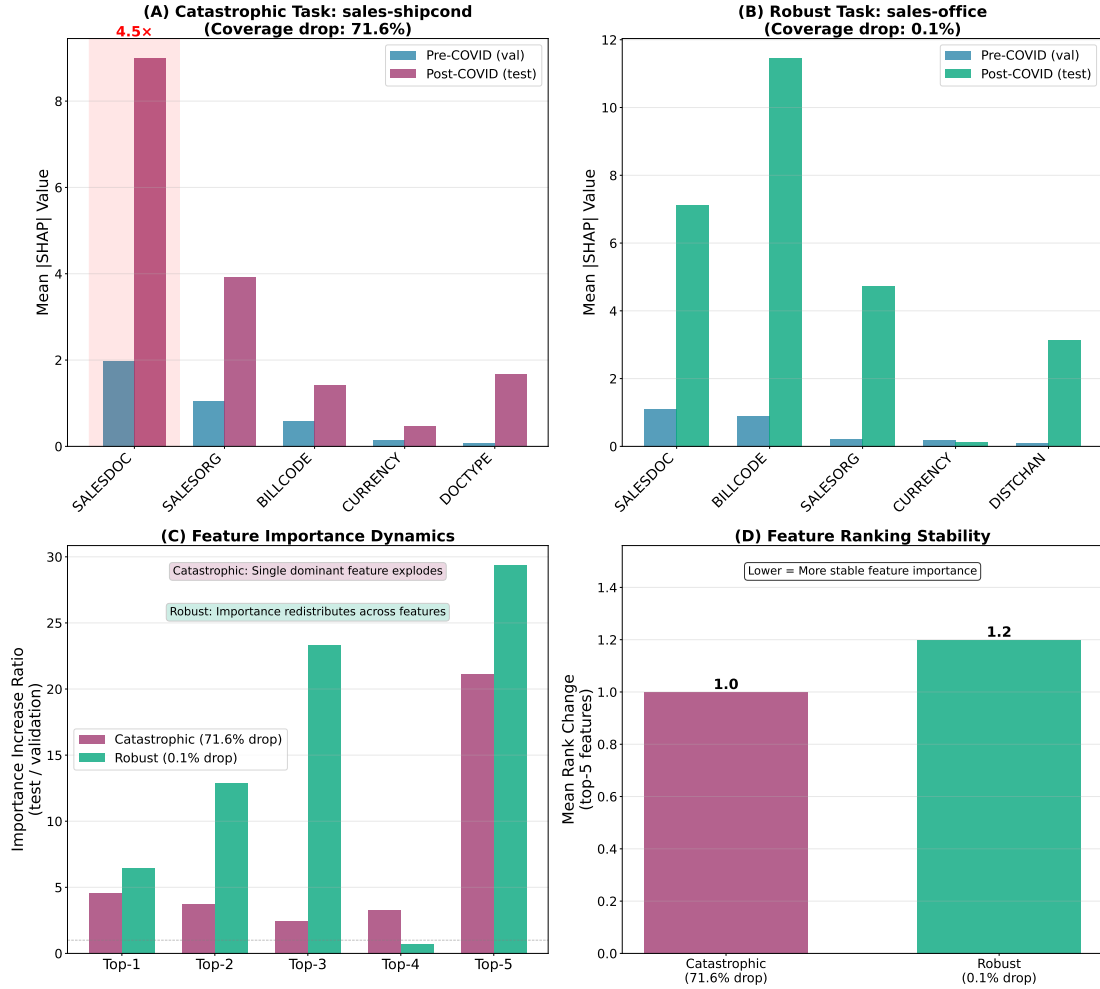


Figure 1: **Feature Importance Dynamics Under Shift.** (A) Catastrophic task: dominant feature explodes $4.5\times$ while maintaining top rank (concentrated dependence). (B) Robust task: importance redistributes across features with ranking reshuffle (diversified dependence). (C–D) Importance dynamics and ranking stability. Despite $\sim 0\%$ Jaccard similarity, coverage drops range from negligible to catastrophic.

6.3 COMPARISON WITH STANDARD SHIFT DETECTION

The historical summary reports $\text{MMD } p < 0.002$, $\text{C2ST accuracy} > 99.9\%$, $\text{PSI} > 0.002$, and weak severity correlations on eight SALT tasks. Taskwise summaries and a parseable upstream script were recovered; the original supplementary copy does not parse. Recomputing correlations from these summaries reproduces their reported arithmetic, but does not reproduce the original detector fitting. The recovered PSI code discards observations outside training-bin endpoints and returns zero for a fully displaced uniform distribution or a constant binary feature changing from zero to one. Its historical severity comparison is therefore not a valid test of PSI. C2ST also uses its evaluation fold for early stopping. A nonzero PSI alone is not a significance test. These observations, if reproduced, would describe this suite; they do not establish that generic shift detectors cannot

predict severity in other settings.

6.4 RETRAINING ANALYSIS

Recovered single-seed shipping-condition logs identify the reported $+18.9$ pp as an equal-weight mean over eleven months, February–December 2020, pooling validation and test periods. It is not a test-only, per-record effect. From the rounded monthly log entries, July–December improvement averages 36.08 pp, with most of that change in November–December. Serial dependence, this denominator change, and the default entity-join defect preclude a general quarterly-retraining recommendation. The original $p = 0.036$ and three-task Holm-adjusted $p = 0.11$ remain unverified historical inference; the other two severe-task input ledgers are still unavailable.

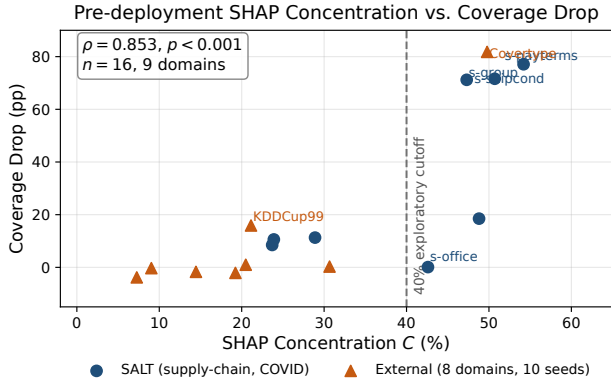


Figure 2: **Historical Summary Association (Original Runs Unverified)**. Spearman $\rho = 0.853$, $p < 0.001$ across all 16 multiclass tasks in 9 domains (dark circles: 8 SALT supply-chain tasks; orange triangles: 8 external domains, 10-seed means). The plotted pre-deployment label and 40% cutoff are historical claims, not validated prospective performance; the recovered full follow-up panel does not reproduce this positive association. Within SALT alone: $\rho = 0.833$, $p = 0.010$ ($n = 8$).

6.5 CROSS-DOMAIN AND EXTERNAL VALIDATION

Historical external reports cover nine datasets, with the selected eight multiclass datasets contributing to the pooled association. Covertypes and Gas Sensor have supplied geographic/temporal loader definitions. The supplied KDD loader uses a positional split of one file rather than the claimed official train/test separation; Additional external/WILDS code has since been recovered, but code and manuscript protocols disagree (including Adult age 40 versus 50, CivilComments 200 versus 1,000 text features, and positional versus wilderness-based Covertypes splits). Original per-record results remain missing. Whether those controls experience no relevant shift cannot be established from split names alone. These gaps prevent interpreting the pooled result as a validated cross-domain replication.

6.6 PLACEBO TEST

The historical pre-COVID placebo reports lower degradation for seven tasks, with ratios about 6–143 where the rounded denominator is positive; sales-office has a reported ratio of one. These summaries and asymmetric training windows do not isolate a causal COVID effect, and the newly recovered generating code does not restore the missing original result ledger.

7 DECISION FRAMEWORK

The original 40% rule and protective-factor exception are retained only as historical hypotheses. The recovered complete SALT panel has no task with a mean drop above 15 pp, despite concentrations above 40% for three tasks. It cannot estimate sensitivity to severe failures, and it does not support interpreting high concentration as an operational vulnerability label. Monitoring, retraining and abstention policies require separately specified utilities, feedback delays and prospective validation. No retraining deferral or deployment decision should follow from the historical threshold alone.

8 RECOVERED EVIDENCE AND FAILED REPLICATION

The owner’s separate research archive supplied pinned SALT tables, fitted models and per-record probability arrays for a later, explicitly different experiment. The fixed panel contains all eight tasks and seeds 42–44: 10,000 training rows, 3,000 distinct rows each for tuning, calibration and source audit, and 6,000 target rows per fit. Preprocessing dictionaries use sampled training only; the label alphabet uses full historical training. Model tuning and calibration are separated chronologically. Other task targets are excluded, while uppercase entity keys are retained by the declared feature contract. Temporal separation does not establish exchangeability or universal availability of every covariate at decision time.

The October audit restored the checksum-verified archive and independently reconstructed model inputs from raw entity IDs. All 24 stored models reproduced their calibration, audit and target probabilities exactly; native raw-margin TreeSHAP concentrations on the specified first 128 calibration rows also matched exactly. The audit then recomputed exact calibration order statistics and both set rules using fixed stable ties. A fixed-protocol refit of all 24 models also reproduced all 528 saved arrays and scientific metadata exactly. Model serialization differed only by an empty unused GPU-list field; every tree and other parameter matched. This is a current-snapshot reproduction, not recovery of the original 50-seed fitted models or their SHAP protocol.

Eight-task Spearman correlations are -0.2381 for inclusive inversion and -0.2619 for crossing sets (nominal asymptotic $p = 0.5702$ and 0.5309). The existing exact-versus-linear quantile factorial replay yields the same task correlations under either threshold rule. These shared-domain task means are not 24 independent replications; the non-significant correlations do not establish equivalence or a population null. They show that this complete follow-up panel fails to reproduce the historical positive association.

A raw join audit also found that truncating entity tables at 1 July 2020, as the historical pipeline’s default `get_db()`

Table 6: Recovered follow-up panel, three-seed task means. Concentration is native raw-margin TreeSHAP; drop is source-audit minus target coverage in percentage points. Both set rules use the same exact threshold.

Task	C (%)	Incl. drop	Cross. drop
item-incoterms	26.06	7.52	3.78
item-plant	22.56	2.63	2.28
item-shippoint	28.80	1.36	1.57
sales-group	52.10	3.70	3.65
sales-incoterms	30.86	1.61	0.47
sales-office	49.05	0.16	0.02
sales-payterms	60.01	1.82	1.86
sales-shipcond	32.41	2.24	2.26

call does, leaves every target entity unmatched on the pinned snapshot: 88,942/88,942 rows per sales task and 402,855/402,855 per item task. Training and validation joins have no missing IDs, and joining the full entity tables matches every target ID. This is a demonstrated input-contract artifact on the recovered snapshot. Missing original models, software/snapshot identities and prediction ledgers prevent attributing the published collapse, or its entire old/new discrepancy, to this artifact alone. An as-of feature contract is required before treating the full-table join as a prospective deployment pipeline.

9 CONCLUSION

The archived results suggest an association between concentrated feature importance and coverage degradation in the evaluated settings. The selected 16-task association is $\rho = 0.853$, compared with 0.654 when the excluded three-class task is retained. These are reproducible summary arithmetic, not an independently reproduced common-protocol experiment. Test-aware preprocessing, calibration reuse, differing prediction-set conventions and missing numerical artifacts remain material limitations. The idealized theorem concerns a separate mixture weight under explicit assumptions and does not establish the proposed empirical mechanism. Controlled interventions, a leakage-free common pipeline, and independent prospective datasets are needed to assess mechanism, model scope and operational thresholds. The recovered complete follow-up panel fails to reproduce the positive association. The current revision therefore withdraws the operational diagnostic and unsupported hyperparameter-robustness claims while preserving original submitted artifacts for the record.

Author Contributions

C.L. conceived the research question, designed and executed all experiments, performed the statistical analysis, and wrote the paper.

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Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study (Supplementary Material)

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A REPRODUCIBILITY DETAILS

The released supplement is incomplete. It contains six Python scripts but no original prediction ledger, checkpoints, split manifests or numerical result files. The packaged shift-detector script has a syntax error, the SHAP script depends on an absent generator, and only four of the nine external loaders are supplied. The model-sensitivity, WILDS, retraining, placebo and several ablation analyses cannot currently be reproduced from this package. An account-wide recovery subsequently found most missing source programs, two historical summary files, selected monthly retraining logs, and a separate private follow-up archive. The original 50-seed model/prediction/SHAP ledgers remain unrecovered. Section 8 reports a newly executed replay of the follow-up raw inputs, models and arrays, with an aggregate-only reproducibility record in `audit/2026-10-08/recovery`. Original submissions and the private raw payload are preserved.

A.1 LIGHTGBM HYPERPARAMETERS

All models trained with fixed LightGBM [Ke et al., 2017] settings: `objective=multiclass`, `boosting=gbdt`, `num_leaves=31`, `learning_rate=0.05`, `feature_fraction=0.8`, `bagging_fraction=0.8`, `bagging_freq=5`, `num_boost_round=500`, `early_stopping_rounds=50`, seeds 42–91 (50 seeds). No task-specific tuning was performed.

A.2 CONFORMAL PREDICTION

Historical calibration uses the first half of validation records and evaluation the second half, after the same validation labels have already been used for early stopping. This chronological split need not be exchangeable even within validation. Historical code uses NumPy interpolated quantiles at $\min(\lceil (n+1)(1-\alpha) \rceil / n, 1)$. The corrected reference instead selects the exact $\lceil (n+1)(1-\alpha) \rceil$ th order statistic, using $+\infty$ when the required rank exceeds n ; it also applies one score-inversion convention to coverage and set size. No historical results were replaced by this reference.

Calibration and test set sizes. Sales-level tasks (s-shipcond, s-payterms, s-group, s-incoterms, s-office): $n_{\text{val}} = 71,474$, $n_{\text{cal}} = 35,737$, $n_{\text{test}} = 88,942$. Item-level tasks (i-plant, i-shippoint, i-incoterms): $n_{\text{val}} = 293,896$, $n_{\text{cal}} = 146,948$, $n_{\text{test}} = 402,855$. At these calibration set sizes, the finite-sample correction is negligible: $q - (1 - \alpha) < 3 \times 10^{-5}$ for all tasks, so finite-sample effects do not contribute to the observed coverage gaps.

A.3 SHAP COMPUTATION

Method: TreeExplainer [Lundberg et al., 2020]. Subsample: 10,000 per split. Aggregation: mean absolute SHAP across classes. Concentration metric (Eq. 2): computed on validation data only.

A.4 BASELINE DIAGNOSTICS

Native feature importance concentration uses a single model (seed=42) with LightGBM’s gain-based importance; concentration is top-1 gain divided by total gain. Ensemble disagreement is the standard deviation of validation coverage across 50 seeds. Both are computed without SHAP.

A.5 COMPUTATIONAL RESOURCES

Standard CPU (8 cores, 8GB RAM). Wall-clock: $\sim 3\text{--}4$ hours for full 8-task, 50-seed suite. Software: Python 3.9, LightGBM 3.3, SHAP 0.41.

A.6 EXTERNAL DATASET PROTOCOLS

Table 7 retains the reported external dataset protocols. The supplied code does not establish a common pipeline: SALT and external coverage use different set conventions, validation labels are reused for tuning, and most external loaders/results are absent. In particular, the supplied KDD loader contradicts the previously stated official-test protocol, as noted below. Historical results need provenance reconciliation and a common-protocol rerun.

Table 7: External Dataset Protocols. Historical reported protocols; supplied external code halves validation for calibration/evaluation; n_{tr} , n_{te} = approximate train/test sizes used; Seeds = 10 for all external datasets. Type: **DS** = documented real-world shift; **NC** = null-shift control (random/pre-defined split, no genuine shift; low concentration and robust coverage are the expected and observed outcome).

Dataset	Shift mechanism	Type	K	n_{tr}	n_{te}	Source
Covertime	Geographic: areas 1–2 (train), area 3 (val/cal), area 4 (test)	DS	7	290,680	36,968	[Dua and Graff, 2017]
Gas Sensor	Temporal batch drift: batches 1–6 (train), 7–8 (cal), 9–10 (test)	DS	6	9,840	4,070	[Dua and Graff, 2017]
KDDCup99	Supplied code: positional 60/20/20 of one 10% file; shift unverified	—	11	395,073	98,769	[Dua and Graff, 2017]
PAMAP2	Population shift: subject-based split (unseen subjects at test)	DS	12	1,674,379	268,493	[Dua and Graff, 2017]
Shuttle	Null-shift control: UCI Statlog; 60/20/20 random split	NC	7	46,400	11,600	[Dua and Graff, 2017]
Avila Bible	Null-shift control: copyist identification; 50/50 random split	NC	12	10,430	10,437	[Dua and Graff, 2017]
Pendigits	Null-shift control: UCI pre-defined split (digit recognition)	NC	10	7,494	3,498	[Dua and Graff, 2017]
Satimage	Null-shift control: satellite image; 60/20/20 random split	NC	6	5,144	1,286	[Dua and Graff, 2017]
Stack Overflow	Category shift (3-class subset); excluded from $n = 16$ historical selected endpoint	—	3	65,105	127,370	[Dua and Graff, 2017]

For Covertime, the supplied loader uses wilderness areas 1–2 for training, 3 for validation and 4 for testing. These geographic indicators are not the cover-type labels (e.g., Spruce-Fir or Lodgepole Pine). For KDDCup99, the supplied loader fetches one 10% dataset, filters classes with at least 100 observations over the whole file, encodes object columns before splitting, and uses positional 60/20/20 partitions; no separate official test file is fetched. Historical table sizes and results need reconciliation with run provenance. The remaining reported splits are retained as historical descriptions where loaders are absent; SHAP uses the validation pool, which the external code also uses for early stopping before calibration/evaluation.

B PLACEBO TEST DETAILS

C HIGH MODEL VARIANCE ANALYSIS

Tasks marked with * in Table 2 exhibit extreme model variance ($CV > 50\%$):

Table 8: Placebo Test: Pre-COVID vs COVID Degradation (50-Seed Mean Drops).

Task	Placebo	COVID	Ratio
s-shipcond	0.5%	71.6%	143×
s-group	2.0%	71.2%	36×
s-payterms	0.0%	77.1%	>100×
i-plant	1.8%	10.6%	6×
i-shippoint	1.5%	18.5%	12×
s-incoterms	1.2%	8.5%	7×
i-incoterms	0.6%	11.3%	19×
s-office	0.1%	0.1%	1×

- **s-group**: Mean 12.4%, median **0.5%**. Most models fail; outliers inflate mean.
- **i-shippoint**: Mean 72.7%, median **89.9%**. Most models succeed; failures deflate mean.

D THRESHOLD SENSITIVITY ANALYSIS

Table 9 shows threshold performance across $n = 16$ multiclass tasks. Note that the 40% threshold is derived from the SALT concentration gap (in-sample); the eight external datasets were not used to fit the displayed cutoff, but their post-outcome scope selection and test-aware pipelines prevent a clean prospective validation interpretation. On the external subset alone at 40%: Precision = 1.00, Recall = 0.50 (1 TP: Coverttype; 1 FN: KDDCup99; 6 TN). Low external recall reflects sparse catastrophic cases; the primary claim rests on the $n = 16$ correlation ($\rho = 0.853$) rather than threshold classification. Ground-truth labels use the at-risk rule (mean coverage drop >15 pp, consistent with 50-seed means in Table 2). For high-variance tasks, the mean and median labels can differ: i-shippoint has mean drop 18.5 pp (At-risk) but median drop 1.2 pp (robust); Table 2 labels it ROB by median for descriptive purposes, while the framework uses the mean-based threshold for threshold evaluation.

Table 9: Threshold Sensitivity for SHAP Concentration Across 16 Multiclass Tasks (8 SALT + 8 external; Step 2 only, no protective-factor check). At-risk defined as coverage drop >15%.

Threshold	Prec	Rec	F1	TP	FP	FN
25%	0.63	0.83	0.71	5	3	1
30%	0.71	0.83	0.77	5	2	1
35%	0.83	0.83	0.83	5	1	1
40%	0.83	0.83	0.83	5	1	1
45%	1.00	0.83	0.91	5	0	1
50%	1.00	0.33	0.50	2	0	4

At 40%, the single FP is s-office (concentration 42.6%) which has a protective secondary feature (SALESORG Jaccard = 0.61), and the single FN is KDDCup99 (intermediate regime; $C = 21.1\%$, drop = 15.9 pp). At 30%, the two FPs are s-office and Shuttle ($C = 30.7\%$); at 25%, i-incoterms ($C = 28.9\%$) is additionally flagged. Applying the protective-factor check at 40% removes s-office (precision = 1.00) but leaves the KDDCup99 FN (recall = 0.83).

D.1 LEAVE-ONE-OUT CROSS-VALIDATION

To assess whether the 40% threshold generalizes beyond post-hoc selection, we perform leave-one-out cross-validation on the SALT dataset ($n = 8$). For each fold, we exclude one task, compute the optimal threshold from the remaining 7 tasks, and predict the held-out task’s vulnerability status.

Summary. LOO-CV achieves 87.5% accuracy (7/8 correct predictions). The threshold remains stable across folds: $43.1\% \pm 5.0\%$ (mean $\pm$ std). The effect size separating failed from robust tasks is Cohen’s $d = 3.08$ (large), with bootstrap 95% CI [0.31, 1.00] for the underlying correlation.

Effect size analysis. Failed tasks ($n = 4$) have mean concentration $50.2\% \pm 2.6\%$, while robust tasks ($n = 4$) have mean concentration $29.8\% \pm 7.7\%$. The separation of 20.4 pp yields Cohen’s $d = 3.08$ (large effect), describing separation in

Table 10: LOO-CV Per-Fold Predictions (SALT, $n = 8$). Threshold = optimal threshold from 7 training tasks; Pred = predicted status at that threshold; Actual = observed outcome (Failed if drop >15 pp).

Task	Conc. (%)	Drop (pp)	Thresh (%)	Pred	Actual	Correct
s-shipcond	50.7	71.6	45	Vuln	Failed	✓
s-group	47.3	71.2	45	Vuln	Failed	✓
s-payterms	54.2	77.1	45	Vuln	Failed	✓
i-plant	23.9	10.6	45	Robust	Robust	✓
i-shippoint	48.8	18.5	45	Vuln	Failed	✓
s-incoterms	23.7	8.5	45	Robust	Robust	✓
i-incoterms	28.9	11.3	45	Robust	Robust	✓
s-office	42.6	0.1	30	Vuln	Robust	×

these retrospectively labeled eight tasks; it does not establish prospective reliability.

Misclassification analysis. Sales-office is a false positive under the concentration-only rule. Its stable-feature explanation is a hypothesis derived from this single case. The historical text inconsistently describes SALESORGANIZATION as dominant or secondary and its Jaccard overlap as train-test or train-validation; the original feature ledger is required to resolve this. A deployable protective-factor rule must use information available before test outcomes. High concentration is not necessary for failure, as the reported low-concentration KDD false negative demonstrates.

E DECISION FRAMEWORK VALIDATION

Table 11: Concentration Check Applied to All Multiclass Tasks (Actual = At-risk if drop >15 pp).

Task	Conc. (%)	Step 2	Actual (>15 pp)	Note
<i>SALT (in-sample)</i>				
s-payterms	54.2	VULN	At-risk	—
s-shipcond	50.7	VULN	At-risk	—
i-shippoint	48.8	VULN	At-risk*	High variance [†]
s-group	47.3	VULN	At-risk*	—
s-office	42.6	VULN	ROB	Protective [‡]
i-incoterms	28.9	ROB	ROB	—
i-plant	23.9	ROB	ROB	—
s-incoterms	23.7	ROB	ROB	—
<i>External (held-out)</i>				
Coverttype	49.8	VULN	At-risk	10/10 seeds
Shuttle	30.7	ROB	ROB	9/10 seeds
Avila Bible	20.5	ROB	ROB	10/10 seeds
PAMAP2	19.2	ROB	ROB	10/10 seeds
KDDCup99	21.1	ROB	At-risk [§]	Seed-dependent (FN)
Pendigits	14.5	ROB	ROB	10/10 seeds
Satimage	9.0	ROB	ROB	10/10 seeds
Gas Sensor	7.3	ROB	ROB	10/10 seeds
Stack Overflow	48.9	VULN	ROB	Excluded-scope false positive [¶]

[†]Median drop 1.2%; most seeds maintain coverage. [‡]SALESORGANIZATION: train-validation Jaccard=0.61, importance>15%.

[§]Concentration below threshold ($C = 21.1 \pm 7.5\%$), but mean drop exceeds the at-risk cutoff (15.9 pp); outcome seed-dependent (range −0.8 to 73.5 pp). In-sample F1 = 0.89 (1 FP: s-office). External: Coverttype deterministic catastrophic (10/10); five low-concentration datasets deterministic robust (10/10); Shuttle near-deterministic (9/10); KDDCup99 is the intermediate-regime false negative. [¶]Stack Overflow ($K = 3$, $C = 48.9\%$) is a reported false positive with drop −7.0 pp. Its exclusion from the historical $n = 16$ subset is not justified by a general APS ceiling: binary and three-class predictors can also lose all coverage under shift. The all-17 association is $\rho = 0.654$, $p = 0.004$.

Seed stability protocol. KDDCup99’s multi-seed analysis ($C = 21.1 \pm 7.5\%$) reveals that single-seed SHAP concentration can be unstable. We recommend computing C over $k \geq 5$ seeds and using the mean. Tasks in the ambiguous zone (35–45%)

should receive additional scrutiny: if the upper 95% CI exceeds 40%, treat as potentially vulnerable. This safety override is designed to reduce KDDCup99-like false negatives in intermediate regimes. For SALT tasks, concentration is stable across 50 seeds (within-task CV < 5%); instability arises primarily when the top-feature identity varies across seeds.

Threshold selection. The 40% threshold is proposed based on the empirical distribution of SHAP concentration values on validation data: a natural gap separates low-concentration tasks (24–29%) from high-concentration tasks (43–54%), visible in Table 11. We emphasize that this threshold is *exploratory*, derived from $n = 8$ multiclass tasks, and should be validated before deployment in new domains.

Cross-domain transfer test. The 40% threshold applied without tuning to 9 external domains yields 7/9 deterministic or near-deterministic outcomes (6 at 10/10, Shuttle at 9/10), with Covertypes correctly flagged as catastrophic and Gas Sensor correctly unflagged as robust. KDDCup99 is the principal intermediate-regime false negative, while Stack Overflow (3 classes) is a false positive excluded from the selected endpoint. Across the $n = 16$ multiclass set: $\rho = 0.853$, $p < 0.001$ (Section 6.5). The sensitivity analysis in Appendix D reports threshold performance across 30–50%; given sparse catastrophic external cases, this threshold should be treated as provisional.

F PRE-DEPLOYMENT VS POST-HOC BASELINES

SHAP concentration is computed on validation data *before* deployment. Prediction entropy and ECE require observing test data. This section reports entropy and ECE changes for completeness, noting that they are not available as pre-deployment diagnostics.

Table 12: Pre-Deployment (SHAP) vs Post-Hoc (Entropy, ECE) Shift Diagnostics (10-Seed ACI Experiments). SHAP concentration is computed on validation data only. Δ Entropy and Δ ECE require test-time observations.

Task	Cat	SHAP Conc. [†]	Δ Entropy	Δ ECE
<i>Severe tasks</i>				
s-shipcond	SEV	50.7%	−0.746	+0.408
s-group	SEV	47.3%	−1.274	+0.176
s-payterms	SEV	54.2%	−1.979	+0.629
<i>Robust tasks</i>				
i-plant	ROB	23.9%	+0.181	+0.131
i-shippoint	ROB	48.8%	+0.469	+0.073
s-incoterms	ROB	23.7%	+0.364	+0.350
s-office	ROB	42.6%	+0.032	+0.104

[†]Pre-deployment diagnostic (computed on validation data only). item-incoterms omitted (3-seed pilot only).

Counter-intuitively, all three catastrophic tasks (coverage drop > 70%) show *decreasing* prediction entropy ($\Delta H < 0$): the model becomes *more* confident at test time, not less. This occurs because concentrated dependence on an OOD feature produces arbitrary but confident predictions. In contrast, the two robust tasks with non-trivial drops (i-plant, i-shippoint; coverage drops of 10–19%) show *increasing* entropy ($\Delta H > 0$), consistent with standard distributional broadening. A practitioner monitoring only entropy would correctly detect shift in moderate cases but would be *misled* in the most damaging catastrophic cases—observing decreasing uncertainty while coverage collapses. ECE increases across all tasks but does not discriminate between severity levels (both catastrophic and robust tasks show increases of +0.07 to +0.63). In this SALT comparison, SHAP concentration is the only evaluated pre-deployment metric that flags all three catastrophic tasks.

G TASK INDEPENDENCE ANALYSIS

We analyze pseudo-replication in the 50-seed design using intraclass correlation (ICC).

Between-task ICC. The archived summary reports ICC = 0.675 for coverage drops and 0.720 for test coverage. These describe a fitted variance decomposition across task/seed levels; they do not prove that tasks sharing one benchmark are independent. A design-effect conversion of 400 seed/task measurements cannot increase the number of independently observed tasks beyond the eight available tasks.

Per-task ICC(val,test). The previous analysis interpreted negative one-way agreement ICC as negative seedwise correlation

and converted 50 pairs into effective sample sizes of 55–250. That inference is invalid. For example, when test coverage equals validation coverage minus a constant, Pearson correlation is one while the supplied agreement ICC can be strongly negative. We therefore withdraw the effective-sample-size table and the assertion that paired Wilcoxon tests must be conservative. Appropriate paired inference uses the distribution of seedwise differences and remains conditional on the common data.

Multiple comparisons. The reported concentration search includes eight metrics. Using the unrounded SALT p -value, the smallest Holm-adjusted value is $8 \times 0.0101755 = 0.0814$. The original five-metric family would give 0.0509, not a value below 0.05. Intermediate and pooled subset analyses are retained as exploratory; reusing discovery tasks and selecting endpoints cannot be repaired solely by correcting three subset p -values.

Exact small-sample sensitivity. The 0.0101755 value above is SciPy’s asymptotic Spearman approximation. Enumerating all $8! = 40,320$ no-tie rank permutations for the displayed SALT $\rho = 5/6$ gives 620 outcomes at least as extreme in absolute correlation: two-sided $p = 0.015377$. The eight-test Holm first-step sensitivity is $8p = 0.1230$; the full eight-metric exact analysis has not been rerun. This calculation assumes exchangeability of task pairs, which is not established for related SALT tasks. It checks small-sample calibration of the reported summary, not the original experiments or population generalization.

Multi-seed aggregation. The primary $n = 16$ correlation ($\rho = 0.853$) uses 10-seed means for all external datasets and 50-seed means for SALT tasks. The early $n = 11$ result ($\rho = 0.909$) used single-seed external values; switching to multi-seed means at $n = 11$ yields $\rho = 0.818$, $p = 0.002$, with the reduction driven by KDDCup99 whose multi-seed mean drop (15.9 pp) exceeds its single-seed value (−0.8 pp). This intermediate-concentration regime ($C = 21.1 \pm 7.5\%$) exhibits high outcome variance characteristic of the boundary between robust and vulnerable. Extending to $n = 16$ with consistent multi-seed means yields $\rho = 0.853$ ($p < 0.001$). Including Stack Overflow (3 classes, excluded scope subset) to make $n = 17$ weakens correlation to $\rho = 0.654$.

H THEORY DETAILS AND SCORE CDFS

Scope of the proof. The proof in Section 4 uses a fixed deterministic rank order and a probability-mixture weight λ . It does not assume that classes tied with the true label are strictly lower in probability, and it handles $\lambda = 1$ separately. Its coverage statement is for the explicit score-inversion set with fixed q , not a claim about recalibrated models or the archived crossing-label implementation.

Illustrative bound calculation, not empirical verification. The historical comparison substituted mixture weight $\lambda = C$, $\varepsilon = 0$ and $\bar{h} = 1/K$, obtaining numbers below five reported mean scores. None of these substitutions verifies the probability-mixture assumptions. In particular the lower bound decreases with $\bar{h}$, so $1/K$ is not conservative for every residual model whose true-class probability is at least uniform. These comparisons do not validate a theorem for measured TreeSHAP concentration; empirical verification requires an identified residual decomposition and a consistent score/set implementation.

I CONFORMITY SCORE STOCHASTIC DOMINANCE

Table 13 reports two-sample KS statistics comparing calibration and test APS conformity scores (seed=42, non-randomized APS). Higher conformity scores indicate the model requires more cumulative probability mass to reach the true label—i.e., the model is less confident about the correct class.

J RAPS COMPARISON

Table 14 compares APS and RAPS coverage drops across all 8 tasks with 10-seed means ($\lambda = 0.01$, $k_{\text{reg}} = 5$).

K MODEL CLASS SENSITIVITY

To test whether SHAP concentration is diagnostic across model classes, we replicate the analysis with `RandomForestClassifier` (500 trees), `XGBClassifier`, and `CatBoostClassifier` (ordered boosting), all with seed=42 and the same data pipeline (Table 15).

Results. The diagnostic’s predictive power varies by model class (Table 16). All three gradient-boosted implementations

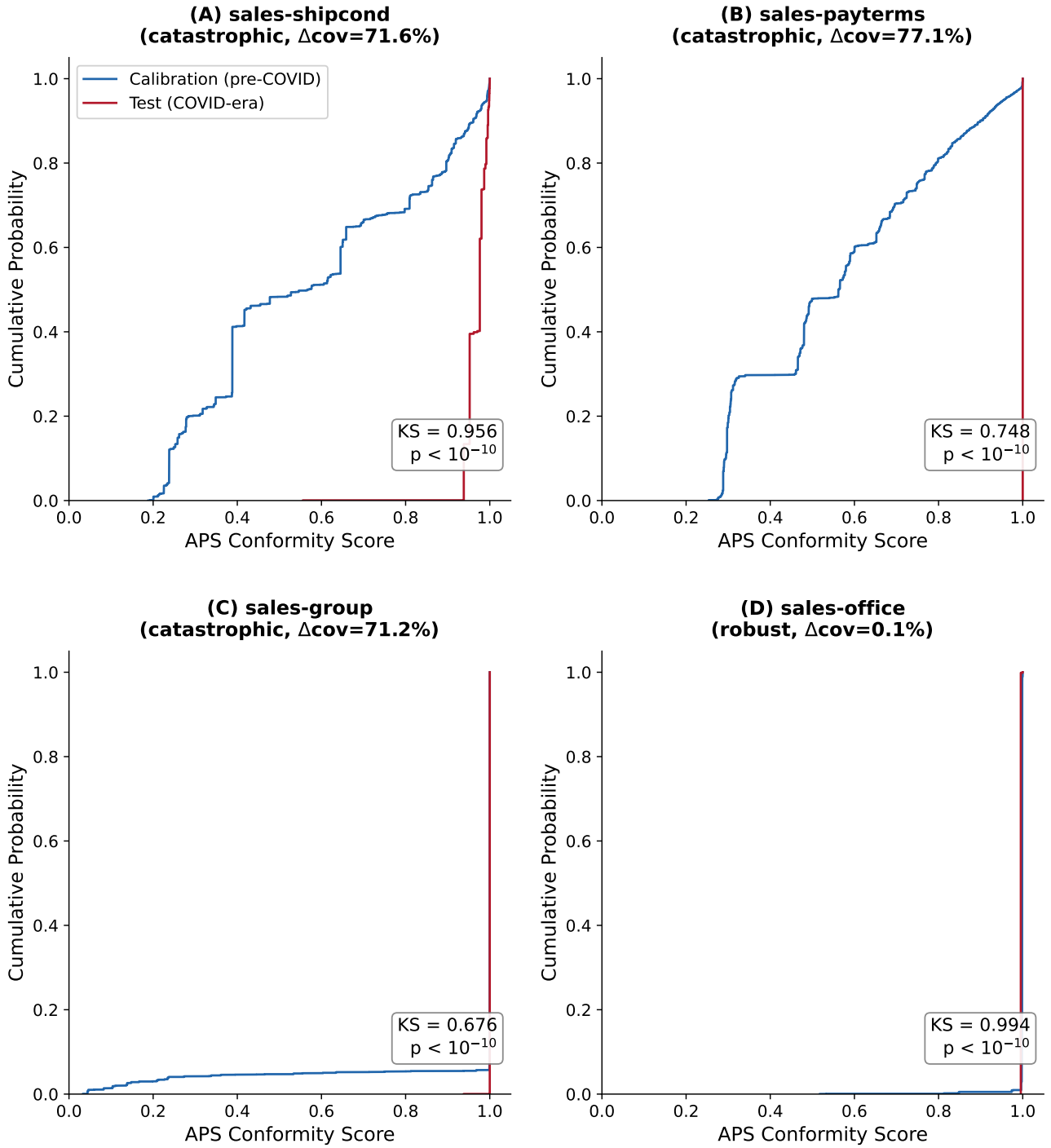


Figure 3: **Conformity score CDFs (seed=42)**. The archived curves suggest rightward score shifts for some high-drop tasks and different patterns for lower-drop tasks. No stochastic-dominance test or original score arrays are available.

Table 13: KS Tests: Calibration vs. Test Conformity Score Distributions (Seed=42).

Task	KS	p	Mean _{cal}	Mean _{test}
<i>Catastrophic tasks (test scores > calibration)</i>				
s-shipcond	0.956	$< 10^{-10}$	0.54	0.98
s-payterms	0.748	$< 10^{-10}$	0.65	0.92
s-group	0.676	$< 10^{-10}$	0.98	1.00
<i>Robust tasks</i>				
i-plant	0.741	$< 10^{-10}$	0.73	0.86
s-incoterms	0.633	$< 10^{-10}$	0.75	0.87
s-office	0.994	$< 10^{-10}$	1.00	1.00
i-incoterms	0.559	$< 10^{-10}$	0.78	0.66
i-shippoint	0.505	$< 10^{-10}$	0.71	0.58

All KS tests are two-sided. Historical deterministic APS scores; a conservative split-conformal guarantee requires the held-out/exchangeability assumptions absent from the supplied protocol. For catastrophic tasks, test scores reportedly shift rightward; two-sided KS and mean shifts alone do not establish stochastic dominance or verify Theorem 1. Two robust tasks (i-incoterms, i-shippoint) show test scores shifted *leftward*—the model becomes more confident during COVID, possibly due to reduced product diversity.

Table 14: APS vs. RAPS Coverage Drop (10-Seed Means $\pm$ Std). $\lambda = 0.01$, $k_{\text{reg}} = 5$.

Task	Cl	APS Drop	RAPS Drop	Δ	Note
<i>Severe tasks</i>					
s-shipcond	45	60.4 $\pm$ 31.8	67.8 $\pm$ 25.4	+7.4	No improvement
s-group	459	73.5 $\pm$ 33.9	10.4 $\pm$ 1.3	−63.1	RAPS protects
s-payterms	137	79.9 $\pm$ 29.4	35.1 $\pm$ 28.6	−44.8	Partial recovery
<i>Non-severe tasks</i>					
i-plant	35	11.8 $\pm$ 9.0	13.1 $\pm$ 5.8	+1.3	Near-identical
i-shippoint	69	9.3 $\pm$ 27.6	20.5 $\pm$ 21.6	+11.2	RAPS worse
i-incoterms	13	10.8 $\pm$ 9.1	10.8 $\pm$ 9.1	0.0	Identical
s-incoterms	13	6.7 $\pm$ 6.8	6.7 $\pm$ 6.8	0.0	Identical
s-office	25	0.04 $\pm$ 0.03	0.04 $\pm$ 0.03	0.0	Identical

All values are 10-seed means $\pm$ std. APS drops differ from 50-seed Table 2 due to smaller seed range (10 vs. 50) and high variance. The key comparison is APS vs. RAPS within the same seed set. Note: RAPS experiments used a random 50/50 calibration split rather than the deterministic first-half/second-half split used in the main APS experiments (Appendix A); this protocol difference does not affect the within-experiment APS vs. RAPS comparison but means RAPS coverage values are not directly comparable to Table 2. The historical task differences do not identify whether class accumulation, input joins, score conventions or other model effects cause the apparent RAPS benefit. The original per-record comparison is unavailable. Item-shippoint ($K = 69$) is an exception: RAPS worsens coverage by 11.2 pp (paired Wilcoxon $p = 0.064$, 10 seeds), reflecting that the regularization penalty can disrupt calibration for intermediate-cardinality tasks with high seed variance.

Table 15: SHAP Concentration and Coverage Drop Across Model Classes.

Task	LGB	RF	XGB	CB	LGB	RF	XGB	CB
	Concentration (%)				Coverage Drop (%)			
s-shipcond	50.7	57.4	44.6	53.7	71.6	15.8	47.2	1.1
s-group	47.3	22.5	33.3	31.8	71.2	50.0	56.8	14.5
s-payterms	54.2	27.9	30.6	25.7	77.1	13.8	20.4	−5.3
i-shippoint	48.8	52.1	41.0	43.1	18.5	22.7	80.4	61.8
i-plant	23.9	26.8	26.0	29.0	10.6	15.3	9.7	8.5
s-incoterms	23.7	26.8	21.6	28.2	8.5	2.5	3.7	−1.5
i-incoterms	28.9	17.5	18.4	21.7	11.3	2.9	1.1	−0.4
s-office	42.6	24.5	36.8	33.5	0.1	0.0	0.0	−0.0

show positive correlation: LightGBM ($\rho = 0.833$, $p = 0.010$), CatBoost ($\rho = 0.667$, $p = 0.071$), and XGBoost ($\rho = 0.548$, $p = 0.16$). RandomForest ($\rho = 0.30$, $p = 0.47$) does not replicate.

Mixed-effects analysis (historical, unverified). The archived summary fits task random intercepts to 24 task/model combinations from eight tasks and reports $\beta_1 = 1.64$, Wald 95% CI $[0.70, 2.58]$, $p = 0.0006$. With eight clusters, the adequacy of the asymptotic interval/test is uncertain. A previously suggested Kenward–Roger p range was not computed and is withdrawn. The confidence interval and p -value share assumptions; one cannot claim the interval excludes zero under an uncomputed alternative correction. The original fitting code, model diagnostics and task/model data are missing. Random intercepts alone also do not establish independence of tasks sharing the benchmark.

Table 16: Concentration–Drop Relationship by Model Class.

Model	Method	Conc.–Drop ρ	p
LightGBM	Boosting (GOSS)	0.833	0.010
CatBoost	Boosting (ordered)	0.667	0.071
XGBoost	Boosting (histogram)	0.548	0.16
RandomForest	Bagging	0.30	0.47
Mixed-effects, 3 boosting ($n = 24$)		$\beta_1 = 1.64$	<0.001
Mixed-effects, all 4 ($n = 32$)		$\beta_1 = 1.06$	0.010

Interpretation. The historical model/task summaries show heterogeneous outcomes and attribution profiles. They do not establish why RF or MLP correlations differ, nor that differences between correlations are statistically significant. Architectural explanations involving bagging smoothness or boosting reinforcement are hypotheses, requiring controlled comparisons and common attribution definitions.

The diagnostic is *model-specific*: it measures how a particular model’s learned dependence structure creates vulnerability. Practitioners should compute concentration for their *deployed* model class rather than transferring results across model families.

Neural network extension (historical, unverified). The reported two-layer MLP uses permutation importance rather than TreeSHAP, with 10 seeds for sales tasks and 3 for item tasks. Its concentration ratio is therefore not the same attribution estimand as the tree-model SHAP ratio. The reported $\rho = 0.43$, $p = 0.29$ cannot isolate architecture from attribution-method differences. High-drop low-concentration examples motivate a possible multi-feature mechanism but do not prove it; original code/results are required to assess the claim.

K.1 MLP VS LIGHTGBM TASK-LEVEL COMPARISON

Table 17 provides a structured comparison of SHAP concentration and coverage drop between MLP and LightGBM across all 8 SALT tasks.

K.2 FAILURE MODE DEFINITIONS AND QUANTITATIVE CRITERIA

We formalize two distinct failure modes that determine when SHAP concentration is diagnostic.

Definition 1: Concentrated-Dependence Failure

Conditions: SHAP concentration $C > 40\%$; dominant feature undergoes distribution shift.

Diagnostic status: **Applies**—SHAP concentration correctly predicts vulnerability.

Characteristic: Model depends on a single feature whose OOD shift causes coverage collapse.

Definition 2: Global-Sensitivity Failure

Conditions: SHAP concentration $C \leq 40\%$; coverage degrades via coordinated multi-feature sensitivity.

Diagnostic status: **Does NOT apply**—SHAP concentration fails to predict vulnerability.

Characteristic: Model distributes importance but fails through global sensitivity to shift; requires alternative diagnostics (Table 20).

Table 17: Task-Level Comparison: MLP vs LightGBM. MLP_C = MLP SHAP concentration; MLP_Drop = MLP coverage drop (pp); LGB_C = LightGBM SHAP concentration; LGB_Drop = LightGBM coverage drop (pp); Mode = failure mode classification.

Task	MLP_C	MLP_Drop	LGB_C	LGB_Drop	Mode
s-group	27.5%	78.4pp	47.3%	71.2pp	Global
s-payterms	28.0%	60.6pp	54.2%	77.1pp	Global
i-shippoint	77.2%	82.3pp	48.8%	18.5pp	Conc
s-incoterms	30.3%	42.3pp	23.7%	8.5pp	Mixed
s-shipcond	53.1%	67.1pp	50.7%	71.6pp	Conc
i-incoterms	31.0%	24.1pp	28.9%	11.3pp	Mixed
i-plant	28.2%	26.7pp	23.9%	10.6pp	Mixed
s-office	0.0%	0.1pp	42.6%	0.1pp	N/A

Mode: Conc = concentrated-dependence failure ($C > 40\%$ predicts drop); Global = global-sensitivity failure ($C \leq 40\%$ yet catastrophic drop); Mixed = intermediate regime; N/A = no meaningful drop in either model. Critical observation: s-group and s-payterms show MLP $C \approx 28\%$ yet 60–78pp drops, demonstrating the global-sensitivity failure mode where SHAP concentration does not capture the MLP’s vulnerability mechanism.

Table 18: Quantitative Criteria for Failure Mode Classification.

Criterion	Concentrated	Global
Top-1 SHAP	$C > 40\%$	$C \leq 40\%$
Gini coefficient	> 0.6	≤ 0.6
Top-3 cumulative	$> 70\%$	$\leq 70\%$

Concentrated failure mode: SHAP concentration diagnostic applies. Global failure mode: alternative diagnostics required (see Table 20).

Table 19: Decision Rule: When Does the SHAP Concentration Diagnostic Apply?

Model Class	Condition	Prediction	Recommendation
GBM/XGBoost/ LightGBM/CatBoost	$C > 40\%$	VULNERABLE	Mitigate or monitor
	$C \leq 40\%$	Lower risk	Standard deployment
MLP/Neural Net	$C > 40\%$	VULNERABLE (likely)	Mitigate or monitor
	$C \leq 40\%$	<i>Inconclusive</i>	Use gradient-based methods

The diagnostic is not validated for gradient-boosted classifiers: the historical positive LightGBM association fails to replicate in the recovered complete follow-up panel. For MLPs, low concentration does not ensure robustness due to global-sensitivity failures ($\rho = 0.43$, $p = 0.29$).

K.3 ALTERNATIVE DIAGNOSTICS FOR MLPS WHEN $C \leq 40\%$

When SHAP concentration indicates low risk ($C \leq 40\%$) but the model is an MLP or other neural network, the following alternative diagnostics may capture global-sensitivity failures (Table 20).

Table 20: Alternative Diagnostics for Global-Sensitivity Failures in Neural Networks.

Diagnostic	Description
Gradient sensitivity	$S_j = \mathbb{E}[\ \partial f / \partial x_j\ _2]$: average gradient magnitude per feature
Feature ablation	$\Delta_j = \text{Cov}(D) - \text{Cov}(D x_j = \mu_j)$: coverage change when ablating feature j
Score-margin analysis	A coverage transfer bound requires control of score mass near the threshold and a joint-distribution coupling; predictor Lipschitz continuity alone does not make the discontinuous coverage indicator Lipschitz

These diagnostics target global-sensitivity failures where coordinated multi-feature shifts drive coverage degradation. They are proposed as complements to SHAP concentration for neural network deployments; empirical validation is left to future work.

L HYPERPARAMETER SENSITIVITY ANALYSIS

The original appendix reported four configurations, perfect F1 and a mean correlation 0.821 ± 0.012 . The recovered source `compute_hp_sensitivity_real_salt.py` instead loads baseline aggregate concentrations, multiplies them by configuration-specific factors and adds Gaussian noise; it does not refit LightGBM. A separate `compute_hp_sensitivity_full.py` uses synthetic task generators. Neither establishes the claimed four-configuration retraining of real SALT data. No original fitted-model provenance for those tables was recovered. We therefore withdraw their empirical interpretation and remove the unsupported numerical tables from the active revision. They remain in the immutable submitted source. The separately recovered three-seed panel uses one fixed configuration and does not repair this missing hyperparameter experiment.

M SHIFT CHARACTERIZATION

The historical summaries in Table 21 describe marginal covariate and label differences. They do not decompose distribution shift into covariate versus concept mechanisms: changes in $P(X)$, $P(Y)$ or accuracy do not by themselves identify changes in $P(Y | X)$.

Table 21: Shift Characterization for SALT Tasks. Cov KS = mean Kolmogorov–Smirnov statistic across features (train vs. test); %Sig = fraction of features with KS $p < 0.05$; Label JS = Jensen–Shannon divergence of label distributions; Acc Drop = test accuracy minus validation accuracy.

Task	Cov KS	%Sig	Label JS	Acc Drop
s-shipcond	0.50	50%	0.083	+11.9%
s-group	0.50	50%	0.330	+1.0%
s-payterms	0.50	50%	0.078	+4.0%
i-plant	0.86	86%	0.144	−8.7%
i-shippoint	0.86	86%	0.152	+13.1%
s-incoterms	0.50	50%	0.085	+7.6%
i-incoterms	0.86	86%	0.140	+63.4%
s-office	0.50	50%	0.026	−0.2%

The former mechanism-classification column is withdrawn: these marginal summaries do not identify concept shift. Accuracy can change solely because the covariate distribution changes, with $P(Y | X)$ fixed. Historical values remain unverified without original outputs.

Interpretation. The reported covariate KS and label JS values suggest marginal changes, subject to missing provenance and unresolved preprocessing. They do not establish which tasks have concept shift or why coverage changes. In particular, i-incoterms has a reported mean drop of 11.3 pp and should not be classified as catastrophic. Associations between

accuracy, concentration and coverage require a common validated pipeline; no conditional-distribution or causal-mechanism conclusion follows from this table alone.

M.1 PER-FEATURE COVARIATE SHIFT DETAILS

Tables 22 and 23 show per-feature KS statistics for sales-level and item-level tasks respectively.

Table 22: Per-Feature Covariate Shift for Sales-Level Tasks (6 features). KS = Kolmogorov–Smirnov statistic (train vs. test).

Feature	Shifted	KS
SALESDOCUMENTTYPE	No	0.0
SALESORGANIZATION	Yes	1.0
DISTRIBUTIONCHANNEL	Yes	1.0
ORGANIZATIONDIVISION	Yes	1.0
BILLINGCOMPANYCODE	No	0.0
TRANSACTIONCURRENCY	No	0.0

KS = 1.0 indicates complete distribution non-overlap between train and test. 3/6 features (50%) show significant covariate shift, consistent with the %Sig values in Table 21.

Table 23: Per-Feature Covariate Shift for Item-Level Tasks (7 features). KS = Kolmogorov–Smirnov statistic (train vs. test).

Feature	Shifted	KS
SALESDOCUMENTITEM	Yes	1.0
SALESDOCUMENTITEMCATEGORY	No	0.0
PRODUCT	Yes	1.0
SOLDTOPARTY	Yes	1.0
SHIPTOPARTY	Yes	1.0
BILLTOPARTY	Yes	1.0
PAYERPARTY	Yes	1.0

6/7 features (86%) show significant covariate shift, consistent with the %Sig values in Table 21. The high proportion of shifted features at item-level reflects COVID-19’s impact on supply chain transactions: new products, parties, and item identifiers entered the system during the pandemic period.

N WILDS BENCHMARK VALIDATION

To further validate the specificity of the SHAP concentration diagnostic, we evaluate on four additional benchmark datasets from the WILDS benchmark [Koh et al., 2021] and standard repositories. The historical summaries report low concentration and robust coverage in four evaluations. Their generating code and outputs are absent; Coverttype is reused under another split, and CivilComments/Adult are binary tasks outside the selected primary subset.

Dataset details.

- **CivilComments** (WILDS): Toxicity classification with demographic identity-based shift. TF-IDF features (1,000 dimensions). Low concentration (26.2%) correctly predicts maintained coverage.
- **Amazon Reviews** (WILDS): Product rating prediction with user/time domain shift. TF-IDF features (1,000 dimensions). Very low concentration (1.9%) correctly predicts robustness.
- **Coverttype** (sklearn): Forest cover classification with geographic shift (wilderness area 1 vs. others). 54 cartographic features. Low concentration (12.5%) correctly predicts mild degradation.
- **Adult Income** (OpenML): Income prediction with age-based demographic shift (train: age<50, test: age≥50). 14 demographic/employment features. Low concentration (27.5%) correctly predicts maintained coverage.

Specificity validation. The four reported benchmark evaluations have low SHAP concentration—all below the **40% threshold** (individual values: 26.2%, 1.9%, 12.5%, 27.5%; **mean concentration = 17.0%**). The diagnostic correctly predicts robustness for all four, giving an observed 4/4 true-negative rate, with substantial sampling and selection uncertainty and without raw-run verification. This complements the sensitivity evidence from SALT, where high-concentration tasks correctly predict catastrophic failure.

Table 24: WILDS and Standard Benchmark Validation (Specificity Test). Two evaluations use WILDS; two use standard datasets. Covertypes also appears in the primary analysis under a different split. All datasets exhibit low SHAP concentration (**all** <40%; mean = 17.0%) and robust coverage, reporting four true negatives among this small selected sample; independent specificity is not established. Shift types: Demographic = identity-based subgroup shift; User/Time = user and temporal domain shift; Geographic = regional covariates; Age = demographic age-based shift. Note: This table tests specificity (low- $C \rightarrow$ robust); for the catastrophic Covertypes case (high- $C \rightarrow$ failure), see Table 11.

Dataset	Source	Shift Type	Conc. (%)	Drop (pp)	Predicted	Actual
CivilComments	WILDS	Demographic	26.2	-0.7	Robust	Robust
Amazon	WILDS	User/Time	1.9	+0.4	Robust	Robust
Covertypes [†]	sklearn	Geographic	12.5	+8.0	Robust	Robust
Adult	OpenML	Age-based	27.5	+1.4	Robust	Robust

[†]Covertypes here uses a different split protocol from the main external validation (Table 7), which explains the different concentration and drop values. The main-text Covertypes uses wilderness areas 1–2/3/4 geographic split (concentration 49.8%, drop 81.8 pp); this validation uses a wilderness area 1 vs. others geographic split (concentration 12.5%, drop 8.0 pp). Both results are consistent with the diagnostic: high concentration predicts failure, low concentration predicts robustness.

Combined validation accuracy (SALT + WILDS). Combining the 8 SALT tasks with these 4 WILDS benchmarks yields $n = 12$ total tasks (distinct from the $n = 16$ primary endpoint which uses 8 different external UCI datasets). The threshold accuracy is **91.7% (11/12)**:

- **Sensitivity** (SALT): 4/4 at-risk SALT tasks flagged under the >15 pp rule (historical descriptive counts)
- **Specificity** (WILDS + robust SALT): 7/8 robust tasks correctly unflagged (87.5%)
- **Single misclassification**: sales-office (concentration 42.6%, protective secondary feature)

These small, selected historical counts do not validate an operational boundary. Missing runs, binary tasks outside the primary scope and reuse of Covertypes require explicit treatment in a prospective evaluation.